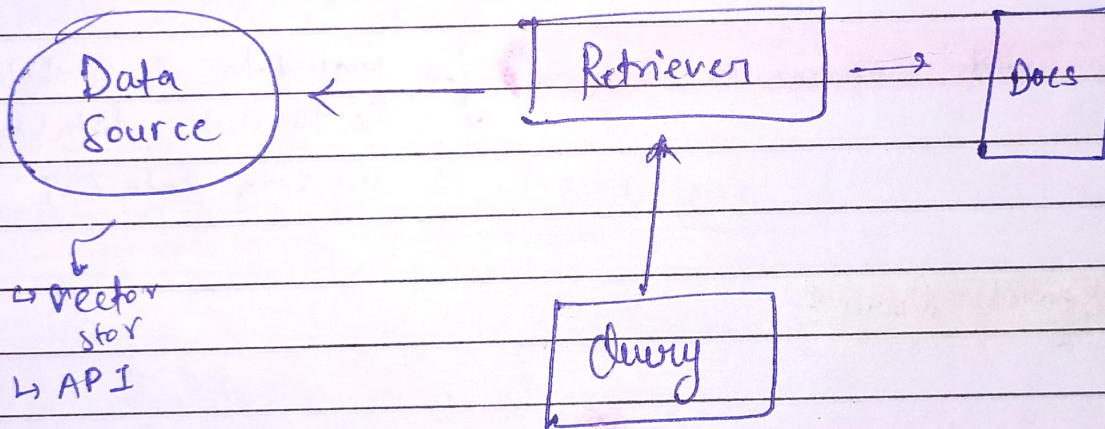


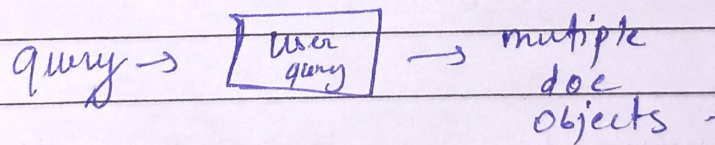
Search engine

Retrievers In LANGCHAIN

- A Retriever is a component in langchain that fetches relevant documents from a data source in response to a User's query.

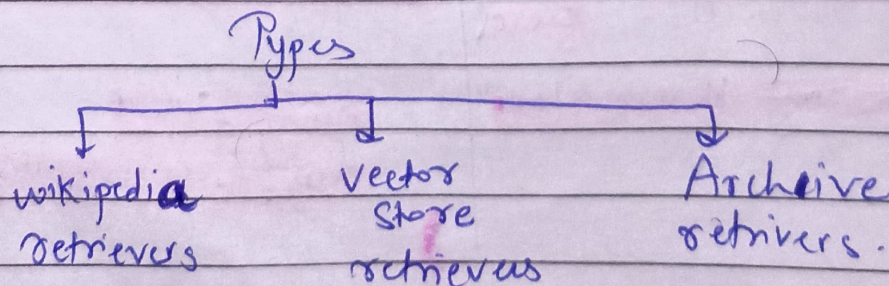


→ Retriever



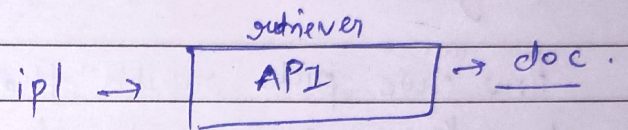
- There are multiple Type of retrievers.

- All retrievers in langchain are runnable.



① wikipedia Retriever :

→ A wikipedia Retriever is a Retriever that queries the wikipedia API to fetch relevant content for a given query.



• How it works :

- 1) you give it a query (like ipl)
- 2) it sends the query to wikipedia's API
- 3) it retrieves the most relevant articles.
- 4) it return them as Langchain Documents objects.

② Vector store Retriever :

↳ A vector store Retriever in langchain is the most common type of retriever that lets you search and fetch documents from a vector store based on semantic similarity using vector embeddings.

• How it works :

1. you store your documents in a vector store (like FAISS, chroma, Weaviate)
2. each document is converted into a dense vector using an embeddings model.
3. when the user enters a query.
 - it is also turned into a vector.
 - The retriever compares the query vectors with the stored vectors.
 - it retrieves the Top-K most similar ones.

③ Maximal Marginal Relevance (MMR) :

- MMR is an Information retrieval algorithm designed to reduce redundancy in the retrieved results while maintaining high relevance to the query.
- How ^{can} we pick results that are not only relevant to the query but also diff- from each other

Why MMR Retriever :-

→ In Regular similarity Search you may get documents that are .

- All very similar to each other.
- Repeating the same info
- Lacking diverse perspectives.

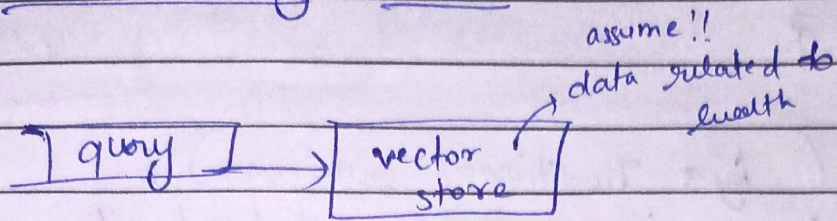
→ MMR Retriever avoids that by

- picking the most relevant document first
- then picking the next most relevant and least similar to already selected docs
- and so on.

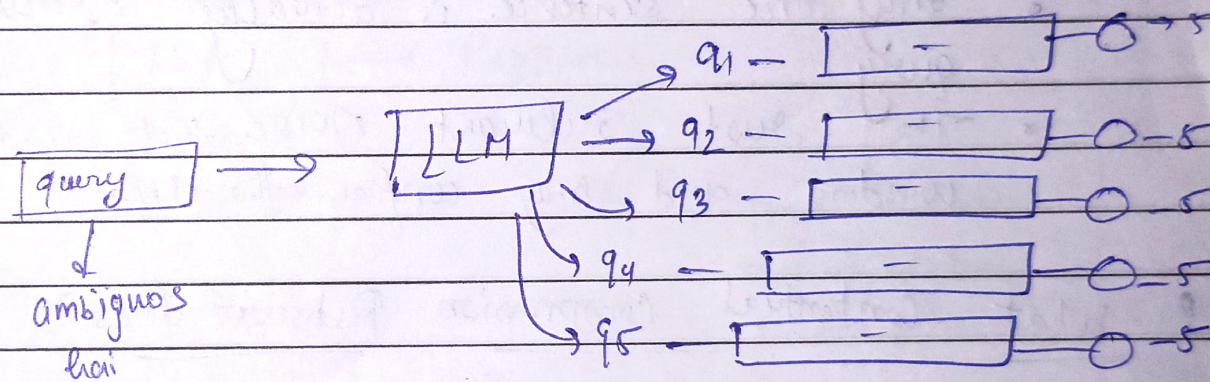
→ This helps especially in RAG pipelines where .

- you want your content window to contain diverse but still relevant information.
- especially useful when documents are semantically overlapping.

④ Multi-Query Retriever :



- query → How can I stay healthy?
- could mean →
 - what should I eat?
 - How often should I exercise?
 - How can I manage stress?



→ sometimes a single Query might not capture all the ways Information is phrased in your Documents -

Code on Github → langchain-Retriever.

⑤ Contentual Compression Retriever :

- The Contentual Compression Retriever in langchain is an Advanced retriever that Improves retrieval Quality by compressing Documents after retrieval. — Keeping only the relevant content based on the user's Query.

• Query :-
what is photosynthesis?

• Retrieved Document (by a traditional retriever) :-
 "The Grand Canyon is a famous natural site.
 photosynthesis is how plants convert light into
 energy.
 Many Tourists visit every year".

• problem :-

- The retriever return the entire ~~passage~~ passage/paragraph.
- only one sentence is actually relevant to the query.
- The rest is just irrelevant noise that wastes content window and may confuse the LLM.

• What Contextual Compression Retriever does?

• Return only the relevant part :-

→ "photosynthesis is how plants convert light into energy."

